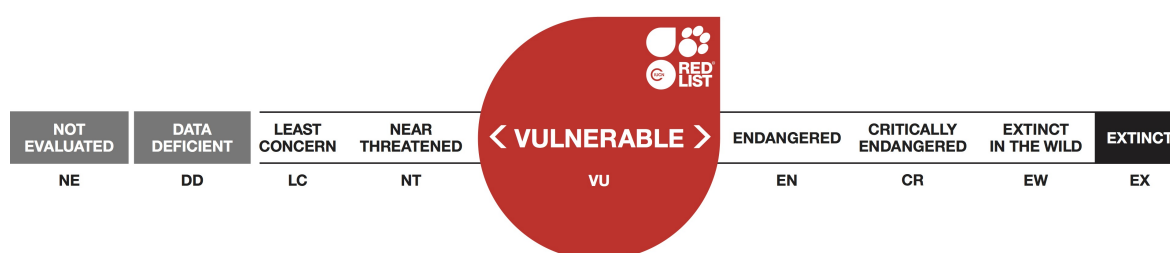


Phelsuma guentheri, Round Island Day Gecko

Errata version

Assessment by: Cole, N., Outlaw, L., Vencatasamy, D. & Mootoocurpen, R.



View on www.iucnredlist.org

Citation: Cole, N., Outlaw, L., Vencatasamy, D. & Mootoocurpen, R. 2018. *Phelsuma guentheri* (errata version published in 2019). The IUCN Red List of Threatened Species 2018: e.T16926A152274946. <http://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T16926A152274946.en>

Copyright: © 2019 International Union for Conservation of Nature and Natural Resources

Reproduction of this publication for educational or other non-commercial purposes is authorized without prior written permission from the copyright holder provided the source is fully acknowledged.

Reproduction of this publication for resale, reposting or other commercial purposes is prohibited without prior written permission from the copyright holder. For further details see [Terms of Use](#).

The IUCN Red List of Threatened Species™ is produced and managed by the [IUCN Global Species Programme](#), the [IUCN Species Survival Commission \(SSC\)](#) and [The IUCN Red List Partnership](#). The IUCN Red List Partners are: [Arizona State University](#); [BirdLife International](#); [Botanic Gardens Conservation International](#); [Conservation International](#); [NatureServe](#); [Royal Botanic Gardens, Kew](#); [Sapienza University of Rome](#); [Texas A&M University](#); and [Zoological Society of London](#).

If you see any errors or have any questions or suggestions on what is shown in this document, please provide us with [feedback](#) so that we can correct or extend the information provided.

Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Reptilia	Squamata	Gekkonidae

Taxon Name: *Phelsuma guentheri* Boulenger, 1885

Common Name(s):

- English: Round Island Day Gecko, Günther's Gecko
- French: Gecko diurne de l'Ile Ronde, Phelsume de l'Ile Ronde
- Spanish: Geco diurno de Guenther

Assessment Information

Red List Category & Criteria: Vulnerable D2 [ver 3.1](#)

Year Published: 2018

Date Assessed: April 23, 2018

Justification:

This species has undergone a severe historical decline that led to its extinction across most of its historical range, and survived for over a century only on an island little over 2 km² in area where native habitat was heavily degraded by the actions of invasive mammalian grazers. Since the 1980s, the eradication of several invasive species, habitat restoration, and active management of this species has led to a stabilization and gradual population increase and the establishment of an additional subpopulation on another restored island within the gecko's historical range. By 2018, the global population was thought to exceed 2,347 mature individuals and was increasing, though subject to periodic fluctuations associated with cyclone damage. The population is predicted to increase more rapidly as additional habitat restoration takes place, particularly of hardwood forest on Round Island. The establishment of an additional subpopulation has increased both the extent of occurrence and area of occupancy (although both remain small, with the area of occupancy presently being less than 8 km² (2.45 km² of actual occupied area)). The species is not subject to major threats on either Round Island or Ile aux Aigrettes at present, although the impacts of ants on both islands and of crows on the latter are unclear.

Invasive species, which regularly invade islands within the Mauritian archipelago (including two recent invasions of Ile aux Aigrettes) and require active control efforts to prevent or eradicate; extreme weather and intense cyclones all represent significant plausible future threats to the survival of this species given its restricted range. Invasive predators pose the greatest threat, being presumed to be the major cause of the species' historical decline. This species is consequently listed as Vulnerable on the basis that it has a very restricted area of occupancy and is subject to multiple plausible future threats, primarily from invasions by exotic predators. The species is, additionally, wholly dependent on ongoing conservation management - particularly invasive species control - to ensure its long-term survival.

Previously Published Red List Assessments

1996 – Endangered (EN)
<http://dx.doi.org/10.2305/IUCN.UK.1996.RLTS.T16926A6602242.en>

1994 – Endangered (E)

1990 – Endangered (E)

1988 – Endangered (E)

1986 – Rare (R)

Geographic Range

Range Description:

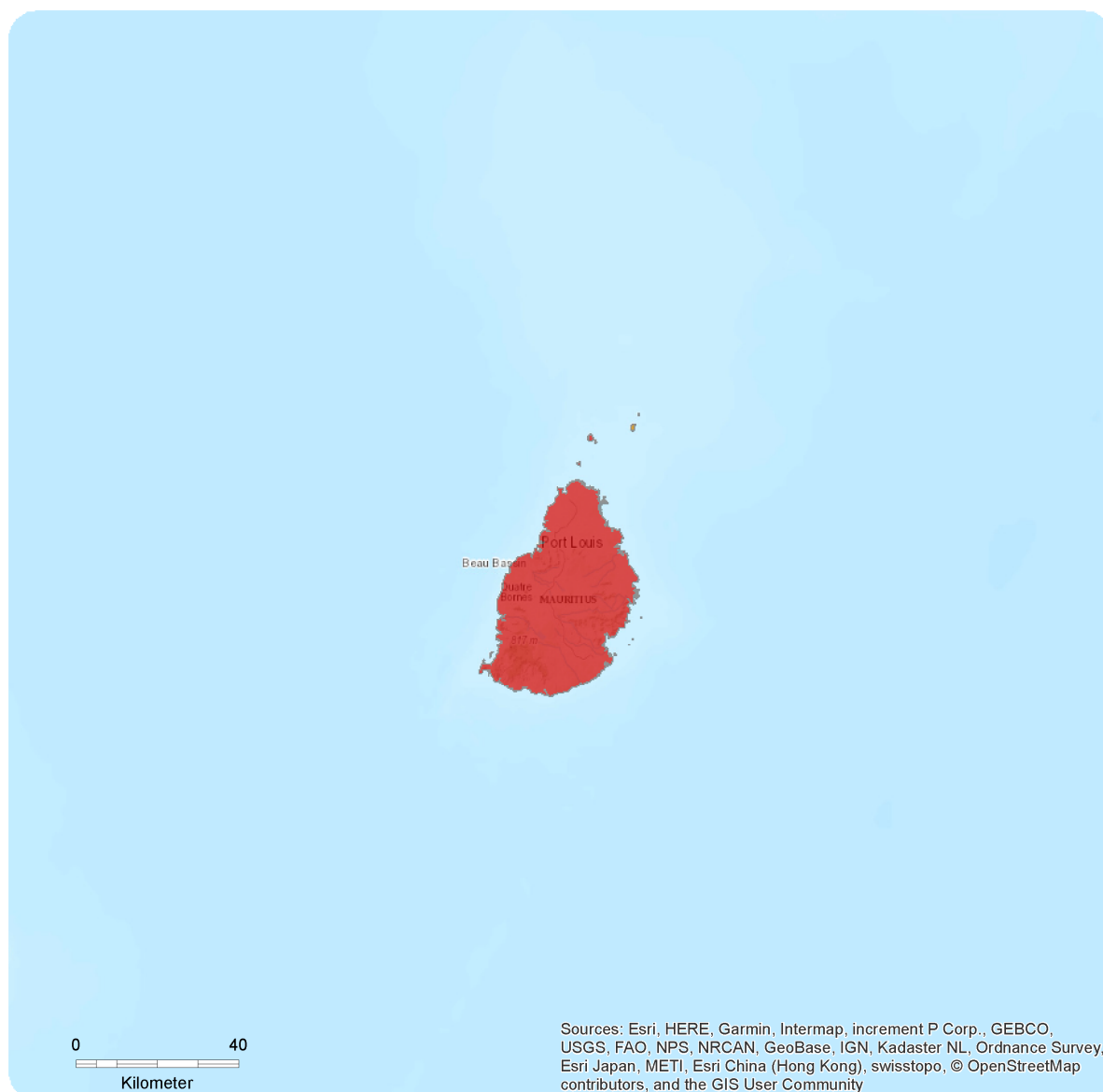
This species is today endemic to Round Island, although bone deposits have been discovered in the southeast and central region of mainland Mauritius and egg scars of this large gecko are still present in the southwest and western regions of mainland Mauritius and on Flat Island (Vinson and Vinson 1969, Arnold 1980, Jones 1988, Cheke and Hume 2008). It is likely that the species was lost from mainland Mauritius soon after its colonization in 1638 and from Flat Island in the 1800s. In 2010, fifty animals were translocated to Ile aux Aigrettes, and are reproducing successfully (Mauritius Reptile Recovery Programme [MRRP] 2010, Cole *et al.* 2014). Although the gecko is suspected to have occurred historically on Ile aux Aigrettes given the proximity of the island to coastal Mauritius where bone deposits were discovered there is presently no evidence that it was native to the island and it is consequently uncertain whether this represents a conservation introduction or a reintroduction (N. Cole pers. comm. June 2012). Of the two islands Round Island has the greatest elevation of 280 m asl., and these geckos occur at all altitudes on the island. The species would have occurred at higher elevations on the Mauritian mainland, given evidence of bone deposits from the central region with an average asl. of 400 m. Following the establishment of the Ile aux Aigrettes subpopulation, the species' area of occupancy is less than 8 km² and the extent of occurrence 70.96 km².

Country Occurrence:

Native: Mauritius

Distribution Map

Phelsuma guentheri



Range

- Extant & Reintroduced (resident)
- Extant (resident)
- Extinct

Compiled by:

International Union for Conservation of Nature (IUCN)



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.



Population

Following the extinction of this species outside Round Island, the remnant population abundance would presumably have declined through the 1800s and much of the 1900s due to extensive vegetation loss on this island caused by introduced goats and rabbits (Cheke and Hume 2008). In 1968, the population size of Günther's Geckos was thought to be between 150 and 300 individuals following the impact of cyclones (Vinson and Vinson 1969). In 1974, the population size was estimated at 2,000 individuals (Temple 1974) and at the start of 1975, the population size was estimated between 1,500 to 1,800 individuals (Vinson 1975). In February 1975, Cyclone Gervaise was thought to have caused a decline of 71% in gecko abundance (Bullock 1986), similar to the declines seen in the 1960s (Vinson and Vinson 1969). The first robust estimate that accounted for detection probability was reported in 1975, of between 437 and 903 individuals within the few remaining palms (Bullock 1986), but this was considered an underestimate by Tonge (1989) as geckos were not recorded within rocky habitats. Being highly cryptic, with no specific peak activity time and being relatively low in abundance, the species is difficult to survey to obtain absolute abundance estimates without extensive effort (Cole *et al.* 2018).

Following the removal of goats in 1979 and rabbits in 1986 (Merton 1987) the palm habitat started to recover (Bullock *et al.* 2002). Nonetheless, surveys in two survey areas on the island conducted at 7-year intervals between 1975 and 1996 failed to detect any change in abundance overall based on encounter rates, although the encounter rate at night was significantly higher following the eradication of rabbits (Bullock *et al.* 2002). In 2002, habitat restoration work started on Round Island with the replanting of endemic hardwoods. In 2007, the abundance of adult geckos was estimated along 65 line transects stratified across seven habitat types, surveyed monthly for a period of one year, in combination with total removal quadrat sampling to account for detection, to obtain an estimate of 2,246 (95% confidence range 1,816–2,676) individuals (Zuël 2009). In 2011, the survey was repeated using distance sampling methodology along 104 line transects stratified across the seven habitat types, and walked day and night within a period of two weeks, to obtain an estimate of 2,308 (95% confidence range 828–3,787) individuals (Cole *et al.* 2018). Whilst the abundance estimates of 2007 and 2011 show negligible change, the encounter rates of geckos within 1 hectare nocturnal survey quadrats in each of the seven habitat types surveyed monthly from 2006 to 2017 exhibit a slight, significant increase in adult abundance (Durrell Wildlife Conservation Trust, Mauritian Wildlife Foundation, National Parks and Conservation Service, unpubl. data).

In 2010, 50 geckos were translocated to Ile aux Aigrettes. Between 2010 and 2017, at least 590 eggs have hatched, and the adult population size was estimated to have increased to approximately 245 individuals by 2013 (Cole *et al.* 2013). The adult population has since fluctuated annually and in 2017 was estimated at approximately 209 adult individuals (MRRP, unpubl. data). The two island subpopulations are 62 km apart with minimal means of natural dispersal.

Current Population Trend: Increasing

Habitat and Ecology (see Appendix for additional information)

This gecko is thought to naturally inhabit humid to dry forest and open mixed woodland dominated by palms and screwpines. Bones deposits and egg scars from mainland Mauritius associated with these vegetation types indicate its historical occurrence in these habitats on the island. On Round Island, its abundance is positively correlated with palm-rich habitat dominated by *Latania loddigesii* palms

(Bullock 1986, Cole *et al.* in press) and *Pandanus* spp. screw pines, but can be found occupying all habitats on the island up to 280 m asl. Despite the species' apparent association with palm habitat and considerable palm recovery following the removal of introduced herbivores, the Round Island subpopulation of geckos has not increased as expected (Cole *et al.* 2018). The introduced population on Ile aux Aigrettes occurs in restored coastal dry forest, which is likely to be a more optimal forest habitat.

Günther's Geckos are one of the largest extant *Phelsuma* species and can reach 28 cm in total length, with males being slightly larger than females, and are highly cryptic (Cole 2009). The visual cells of Günther's Gecko are intermediate between diurnal and nocturnal gecko species (Röll 2000) and as such the gecko exhibits no discrete activity period, being active both day and night (Carpenter *et al.* 2003, Cole *et al.* 2018). Günther's Geckos feed mostly upon invertebrates, but also fruits, nectar and tree resins, other reptiles (including juvenile conspecifics), and small birds, and are themselves preyed upon by the endemic Telfair's Skink (*Leiolopisma telfairii*) and Keel-scaled Boa (*Casarea dussumieri*) (Cole 2009, Züel 2009). The gecko is predominantly arboreal (Bullock 1986, North *et al.* 1994, Carpenter *et al.* 2003), although animals are also frequently encountered in open areas of bedrock on Round Island (Tonge 1989, Cole 2009).

The reproductive season is usually between July and April, with females depositing two eggs per clutch, with up to three clutches (up to four in captivity) laid between September and March taking approximately 68 days to incubate (Bloxam and Vokins 1978, Cole *et al.* 2014). Like all Mascarene day geckos, Günther's Geckos glue their eggs to the substrate. On Ile aux Aigrettes females lay eggs on smooth hardwood species, which are affixed by oviduct secretions to the surface of the wood. On Round Island eggs are laid on palm fronds and in rocky coves, and to a lesser degree on the trunks of screw pines (Vinson and Vinson 1969, Carpenter *et al.* 2003), while on Ile aux Aigrettes females preferentially lay eggs in coastal hardwoods. Approximately 90% of eggs that are deposited on hardwood trunks hatch on Ile aux Aigrettes, which is similar to the survival rate on rock coves in Round Island; survival rates are considerably lower for eggs laid in palms and screw pines (Cole *et al.* 2013, 2014). These hardwood species were once known to have occurred on Round Island and are currently being reintroduced to the island, which is predicted to enhance the reproductive success and population growth of Günther's Gecko (Cole *et al.* 2018). Hatchlings reach sexual maturity at approximately 18 months, and reproduce within the second breeding season post-hatching (MRRP unpubl. data).

The average home-range sizes of males and females within palm-rich habitats on Round Island was estimated at 73 m² and 31 m², respectively (Gerner 2008). On Ile aux Aigrettes, the average home range sizes for male and female geckos within native coastal forest, two years post-translocation, was estimated at 1,799 m² and 374 m², respectively (M. Wynn and N. Cole unpubl. data). The results of repeated surveys on Ile aux Aigrettes - in which individual geckos were identified from their scars, missing toes and tail breaks - estimated an annual apparent survival estimate of 78% (MRRP 2011).

Systems: Terrestrial

Use and Trade

Günther's Geckos, as with all *Phelsuma* species, are listed in Appendix II by the Convention on International Trade in Endangered Species (CITES) of Wild Fauna and Flora. Whilst there may be demand for the species within the pet trade, no legally exported individuals currently exist in captivity.

Threats (see Appendix for additional information)

Historically, the Günther's Gecko population suffered a catastrophic decline, with bone deposits and egg scars indicating that the species was once present on the Mauritian mainland and associated islands. Although the species' presence on this island is not documented in historical accounts, declines and extinction is presumed to be the result of human visitation from the 1500s and occupation from 1638. This led to the introduction of non-native species and widespread habitat destruction and was responsible for the extinction of multiple other species from Mauritius, including other reptiles that survived only on Round Island (which remained free of invasive mammalian predators). Introduced mammalian predators, such as rats and cats are thought to have been the main cause of initial declines and loss (Cheke and Hume 2008). On Round Island, the remnant population of this arboreal gecko would presumably have suffered significant declines through the action of introduced goats and rabbits, which denuded the island of much of its native vegetation prior to the eradication of these exotic grazers eradicated in 1986. The introduced herbivores caused the loss of the hardwoods from Round Island, thus limiting suitable nest site availability for the gecko (Cole *et al.* 2018). This would have both removed optimal habitat for the lizard and exposed the geckos and their eggs to the detrimental effects of severe weather and cyclonic events (Vinson and Vinson 1969, Bullock and North 1984, Bullock 1986, Cole *et al.* 2014), the latter of which has been documented to cause periodic population declines on Round Island (Vinson and Vinson 1969, Bullock 1986). Since around 2002 there has been substantial replanting of lowland forest vegetation on Round Island, and this is likely to improve the species' ability to survive future stochastic events.

The rate of offshore island invasion events in the Mauritian archipelago is approximately three non-native fauna incursions per year, several of which have posed a significant threat to terrestrial biodiversity (Cole and Payne 2015). Since 2015, introduced Indian House Crows (*Corvus splendens*) have spread to the southeast of Mauritius, visiting Ile aux Aigrettes daily, and these are considered a threat to the translocated Günther's Gecko subpopulation by predating juveniles and adults. The Big-headed Ant (*Pheidole megacephala*) predated juvenile and adult geckos, although the extent of their impact is currently unknown on both islands. There is a risk of further ant invasions, such as the tropical fire ant (*Solenopsis geminata*), which is known to predate Mauritian reptiles (Cole 2005). In most cases staff of the Durrell Wildlife Conservation Trust, Mauritian Wildlife Foundation and National Parks and Conservation Service through the MRRP have prevented non-native species incursions, however, the risk of species arriving where there is no current method of removal is high (Cole and Payne 2015).

Conservation Actions (see Appendix for additional information)

Günther's Gecko is part of a current conservation programme by the Durrell Wildlife Conservation Trust, Mauritian Wildlife Foundation, and National Parks and Conservation Service to restore the reptile communities on the offshore islands of Mauritius (Cole *et al.* 2009, 2014). The removal of goats and rabbits from Round Island, followed by ongoing recovery and restoration of vegetation that encouraged the stabilization and growth of the gecko population, allowed the translocation of individuals without threatening the source population (MRRP 2010). The removal of rats and cats from Ile aux Aigrettes by 1991 and restoration of the island's native vegetation (Jones and Hartley 1995, Jones 2008) opened the island for the translocation of Günther's Geckos in 2010, where a subpopulation has since successfully established (MRRP 2010, Cole *et al.* 2014).

Ile aux Aigrettes Nature Reserve continues to be a primary restoration project of the Mauritian Wildlife

Foundation, who have a permanent staff presence on the island, with controlled and guided visitation for the public (Jones 2008). Another key restoration site is Round Island, a Closed Island Nature Reserve where public access is prohibited, which is further protected by the permanent presence of staff and wardens jointly managed by the Durrell Wildlife Conservation Trust, Mauritian Wildlife Foundation, and National Parks and Conservation Service (Jones 2008). Ongoing activities to restore the plant community on Round Island, particularly the reintroduction of native hardwoods, is likely to improve habitat and nesting sites for Günther's Gecko (Cole *et al.* 2018). Biosecurity measures are in place for both islands to reduce and detect non-native species incursions. Recent invasions of Ile aux Aigrettes by feral dogs (*Canis familiaris*) and Tailless Tenrecs (*Tenrec ecaudatus*), the latter of which has not successfully been eradicated (Goder *et al.* 2015), are not known to have impacted this species but highlight the vulnerability of this island to species invasion.

In the next decade it may be possible to reintroduce Günther's Geckos to other islands, such as the Closed Island Nature Reserve Gunner's Quoin (0.70 km²) to the north of Mauritius where the reptile community is being rebuilt (Cole *et al.* 2009, 2014, Goder *et al.* 2015) and the plant community is being restored. It is also likely that as other islands of conservation importance, such as Flat Island (2.53 km²) and Gabriel Island (0.42 km²), start to be restored and invasive species removed, it may be possible to reintroduce Günther's Geckos. The survival of this Mauritian reptile is dependent upon ongoing conservation activities, and the restoration and protection of its habitats.

Credits

Assessor(s): Cole, N., Outlaw, L., Vencatasamy, D. & Mootoocurpen, R.

Reviewer(s): Bowles, P.

Bibliography

- Arnold, E.N. 1980. Recently extinct reptile populations from Mauritius and Réunion, Indian Ocean. *Journal of Zoology* 191: 33-47.
- Bloxam, Q. and Vokins, M. 1978. Breeding and maintenance of *Phelsuma guentheri* (Boulenger, 1885) at the Jersey Zoological Park. *Dodo - Journal of the Wildlife Preservation Trusts* 15: 82-91.
- Bullock, D. and North, S. 1984. Round Island in 1982. *Oryx* 18: 1.
- Bullock, D.J. 1986. The ecology and conservation of reptiles on Round Island and Gunner's Quoin, Mauritius. *Biological Conservation* 37: 135-156.
- Carpenter, A.I., Côté, I.M. and Jones, C.G. 2003. Habitat use, egg laying sites and activity patterns of an endangered Mauritian gecko (*Phelsuma guentheri*). *The Herpetological Journal* 13: 155-157.
- Cheke, A.S. and Hume, J.P. 2008. *Lost Land of the Dodo: and ecological history of Mauritius, Réunion and Rodrigues*. Yale University Press.
- Cole, N. 2005. The ecological impact of the invasive house gecko *Hemidactylus frenatus* upon endemic Mauritian geckos. PhD thesis. University of Bristol.
- Cole, N. 2009. *A Field Guide to the Reptiles and Amphibians of Mauritius*. Mauritius Wildlife Foundation, Vacoas, Mauritius.
- Cole, N. and Payne, C. 2015. *Gongylomorphus bojerii*. *The IUCN Red List of Threatened Species* 2015: e.T62251A13482733. DOI: [dx.doi.org/10.2305](https://doi.org/10.2305).
- Cole, N., Goder, M., Vencatasamy, D., Mootoocurpen, R., Havery, S., Gamble, F. and Nundlaur, V. 2013. Restoration of Island Ecosystems in Mauritius: The Mauritius Reptile Recovery Programme Annual Report 2013. Durrell Wildlife Conservation Trust, Jersey, Channel Islands.
- Cole, N., Goder, M., Vencatasamy, D., Mootoocurpen, R., Le Flohic Gillies, A., Herbert, S., Gamble, F., Ramen, B., Ramjeeawon, D. and Nundlaur, V. 2014. Restoration of Island Ecosystems in Mauritius: The Mauritius Reptile Recovery Programme Annual Report 2014. Durrell Wildlife Conservation Trust, Jersey, Channel Islands.
- Cole, N., Jones, C., Buckland, S., Jhumka, Z., Mootoocurpen, R., Tatayah, V., Bachraz, V., Nundlaur, V., Roopa, P. and Seewajee, P. 2009. The reintroduction of endangered Mauritian reptiles. Mauritian Wildlife Foundation. Vacoas, Mauritius.
- Cole, N., Mootoocurpen, R. and Nundlaur, V. 2018. Relative density estimates of Round Island's reptiles. *The Journal of the Royal Society of Arts and Sciences of Mauritius* 1: 1-22.
- Gerner, T. 2008. Home range, habitat use and social behaviour of the endangered Mauritian gecko *Phelsuma guentheri*. Master's Thesis. Institute of Environmental Sciences, University of Zurich.
- Goder, M., Mootoocurpen, R., Mellier, S., Hayes, R., Nundlaur, V., Roopa, P. and Cole, N. 2015. Restoration of Island Ecosystems in Mauritius: The Reptile Recovery Programme Annual Report 2015. Mauritian Wildlife Foundation, Vacoas, Mauritius.
- IUCN. 2018. The IUCN Red List of Threatened Species. Version 2018-2. Available at: www.iucnredlist.org. (Accessed: 15 November 2018).
- IUCN. 2019. The IUCN Red List of Threatened Species. Version 2019-2. Available at: www.iucnredlist.org. (Accessed: 04 July 2019).

Jones, C.G. 1988. Remains of Gunther's gecko eggs found in the Black River Gorges. *Proceedings of the Royal Society of Arts and Sciences of Mauritius* 5: 135-136.

Jones, C.G. 2008. Practical conservation on Mauritius and Rodrigues: steps towards the restoration of devastated ecosystems. In: A. Cheke and J. Hume (eds), *Lost Land of the Dodo: An Ecological History of Mauritius, Réunion and Rodrigues*, pp. 226-259. A&C Black, London.

Jones, C.G. and Hartley, J. 1995. A conservation project on Mauritius and Rodrigues: An overview and bibliography. *Dodo - Journal of the Wildlife Preservation Trusts* 31: 40-65.

Merton, D.V. 1987. Eradication of rabbits from Round Island, Mauritius: a conservation success story. *Dodo: Journal of the Jersey Wildlife Preservation Trust* 24: 19-43.

MRRP. 2010. Restoration of Island Ecosystems in Mauritius: The Mauritius Reptile Recovery Project Annual Report, June 2010. Durrell Wildlife Conservation Trust, Jersey, Channel Islands.

MRRP. 2011. Restoration of Island Ecosystems in Mauritius: The Mauritius Reptile Recovery Programme Report, 2011 January-June. Durrell Wildlife Conservation Trust, Jersey, Channel Islands.

North, S.G., Bullock, D.J. and Dulloo, M.E. 1994. Changes in the vegetation and reptile populations on Round Island, Mauritius, following eradication of rabbits. *Biological Conservation* 67: 21-28.

Röll, B. 2000. Gecko vision - visual cells, evolution, and ecological constraints. *Journal of Neurocytology* 29: 471-484.

Temple, S. 1974. Last chance to save Round Island. *Wildlife* 16: 370-374.

Tonge, S.J. 1989. A preliminary account of changes in reptile populations on Round Island following the eradication of rabbits. *Dodo - Journal of the Wildlife Preservation Trusts* 26: 8-17.

Vinson, J. 1975. Notes on the reptiles of Round Island. *The Mauritius Institute Bulletin* 8: 49-67.

Vinson, J. and Vinson, J.M. 1969. The saurian fauna of the Mascarene Islands. *The Mauritius Institute Bulletin*, VI: 203-320.

Zuël, N. 2009. Ecology and Conservation of an Endangered Reptile Community on Round Island, Mauritius. PhD Thesis. Universität Zürich.

Citation

Cole, N., Outlaw, L., Vencatasamy, D. & Mootoocurpen, R. 2018. *Phelsuma guentheri* (errata version published in 2019). The IUCN Red List of Threatened Species 2018: e.T16926A152274946.
<http://dx.doi.org/10.2305/IUCN.UK.2018-2.RLTS.T16926A152274946.en>

Disclaimer

To make use of this information, please check the [Terms of Use](#).

External Resources

For [Images and External Links to Additional Information](#), please see the [Red List website](#).

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.5. Forest - Subtropical/Tropical Dry	Resident	Suitable	Yes
3. Shrubland -> 3.5. Shrubland - Subtropical/Tropical Dry	Resident	Suitable	-
13. Marine Coastal/Supratidal -> 13.1. Marine Coastal/Supratidal - Sea Cliffs and Rocky Offshore Islands	Resident	Suitable	-

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
11. Climate change & severe weather -> 11.2. Droughts	Future	Whole (>90%)	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
11. Climate change & severe weather -> 11.4. Storms & flooding	Past, likely to return	Whole (>90%)	Causing/could cause fluctuations	Past impact
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Oryctolagus cuniculus</i>)	Past, unlikely to return	Whole (>90%)	Unknown	Past impact
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Rattus rattus</i>)	Past, likely to return	Whole (>90%)	Rapid declines	Past impact
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Rattus norvegicus</i>)	Past, likely to return	Whole (>90%)	Rapid declines	Past impact
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (<i>Pheidole megacephala</i>)	Ongoing	Whole (>90%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		

8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Solenopsis geminata)	Ongoing	Whole (>90%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Corvus splendens)	Ongoing	Minority (50%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Felis catus)	Past, unlikely to return	Whole (>90%)	Unknown	Past impact
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Capra hircus)	Past, unlikely to return	Whole (>90%)	Unknown	Past impact
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions in Place
In-Place Research, Monitoring and Planning
Action Recovery plan: Yes
Systematic monitoring scheme: Yes
In-Place Land/Water Protection and Management
Conservation sites identified: Yes, over entire range
Occur in at least one PA: Yes
Percentage of population protected by PAs (0-100): 91-100
Invasive species control or prevention: Yes
In-Place Species Management
Successfully reintroduced or introduced benignly: Yes
In-Place Education
Included in international legislation: Yes
Subject to any international management/trade controls: Yes

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Actions Needed
2. Land/water management -> 2.2. Invasive/problematic species control
2. Land/water management -> 2.3. Habitat & natural process restoration
3. Species management -> 3.3. Species re-introduction -> 3.3.1. Reintroduction

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
3. Monitoring -> 3.1. Population trends
3. Monitoring -> 3.4. Habitat trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km ²): 8
Estimated extent of occurrence (EOO) (km ²): 70.96
Number of Locations: 2
Lower elevation limit (m): 0
Upper elevation limit (m): 280
Population
Number of mature individuals: 2347
Continuing decline of mature individuals: No
Extreme fluctuations: No
Population severely fragmented: No
Habitats and Ecology
Generation Length (years): 5.9

Errata

Errata reason: A minor typographical error has been corrected under the Rationale.

The IUCN Red List Partnership



The IUCN Red List of Threatened Species™ is produced and managed by the [IUCN Global Species Programme](#), the [IUCN Species Survival Commission](#) (SSC) and [The IUCN Red List Partnership](#).

The IUCN Red List Partners are: [Arizona State University](#); [BirdLife International](#); [Botanic Gardens Conservation International](#); [Conservation International](#); [NatureServe](#); [Royal Botanic Gardens, Kew](#); [Sapienza University of Rome](#); [Texas A&M University](#); and [Zoological Society of London](#).